

SDE

AMAZON · ORIGIN



3 YRS · JAVA · DISTRIBUTED · AWS

SR·E

NEXT ROLE · DEST

PASSENGER

PRAKHAR SHEKHAR PARTHASARTHI

FULL-STACK SOFTWARE ENGINEER · EX-AMAZON



SEAT 1A

P90 LATENCY

47%

cut · booking

CONVERSION

+33%

traveler lift

GET · REDIS

195k

rps · from scratch

ON-CALL

20→5

tickets / shift

ITINERARY

SUMMARY

Full-stack engineer with **3+ years at Amazon** (Payments & Travel) — **owned 4 microservices end-to-end** through design, build, deploy, and on-call. Java-heavy backend (**Java 17 / Spring Boot**, `CompletableFuture`, bounded `ExecutorService`), idempotent event-driven systems over **SQS + Step Functions**, typed infra in **AWS CDK (Java)**, paired with **React (SSR & CSR) + Redux**. Reliability-first: **47% p90** cut, **+33% conversion**, on-call **20+→~5** per shift.

SEGMENTS

EXPERIENCE

Software Development Engineer Amazon · Travel (Hotels & Flights)

JUL 2022 – SEP 2025 · BENGALURU

- **Service ownership.** Owned **4 customer-facing microservices** (search, listing, room selection, order review) on Spring Boot/MVC — REST controllers, service logic, `@Repository` DynamoDB DAOs via Spring DI.
- **JVM concurrency.** Refactored sequential downstream calls into `CompletableFuture.allOf()` on a bounded `ExecutorService`; parallelised 6 partner-API lookups — hotel landing p90 **1.1s→0.59s (-47%)**, conversion **+33%**.
- **Event-driven.** Idempotent Java webhook for real-time flight schedule changes — SQS ingestion, DynamoDB dedup keys, audit logging — highest-impact pain category **-43 bps CPT**. Step Functions booking state machine.
- **IaC — AWS CDK (Java).** Authored all service infra (Lambda, DynamoDB, SQS, Step Functions, API Gateway, CloudWatch). Migrated 2 services off hand-rolled CloudFormation — **halved** rollback incidents.
- **Ops excellence.** Mined multi-year on-call corpus — tickets **20+ → ~5 / shift** (~2 SDE-weeks/week reclaimed); resolved **150+** incidents; **80%+** test coverage (JUnit 5 + Mockito).

Software Development Engineer, Intern Amazon

MAY 2021 – JUL 2021 · FEB 2022 – JUL 2022 · BENGALURU

- Migrated legacy GraphQL resolvers to **Java** — rebuilt resolver classes wired to DynamoDB via Spring DI, exposed through API Gateway + Lambda.
- Shipped **One-Click Renewal** for auto-insurance (async Java pre-fill); applied OO design + Google Guice DI with **90%+** coverage.



CABIN

CORE TECHNICAL STACK



JAVA & JVM

Java 17, Spring Boot, Spring MVC/DI, Google Guice, JUnit 5, Mockito, Maven, Gradle · [CompletableFuture](#), [ExecutorService](#), GC tuning



AWS

Lambda (Java), DynamoDB, SQS, Step Functions, API Gateway, AppSync, CloudWatch, ECS, **CDK (Java)**



DATA & STORAGE

DynamoDB, Redis, NoSQL modelling, SQL, caching strategies



DISTRIBUTED SYSTEMS

Microservices, REST, GraphQL, event-driven, idempotency & dedup, state machines, exponential-backoff retry, DLQ, at-least-once, TDD



FRONT-END

React (**SSR & CSR**), Redux (1.5 yrs prod), HTML/CSS, JavaScript/TypeScript



TOOLING & OTHER

Git, CI/CD, JaCoCo gates, Agile/Scrum, on-call · Python, C++, VTL

DUTY-FREE

SELECTED PROJECTS

Redis Server in Java

github.com/praxstack/redis-server-java

RESP2-compatible Redis from scratch. Thread-per-client on a bounded [ExecutorService](#); lock-free reads via [ConcurrentHashMap](#); hybrid TTL eviction; graceful shutdown; TCP integration tests.

Bench (100 clients): SET 154k · GET 195k · INCR 128k rps · p50 < 0.3ms · 42 tests, 100%.

Markdown Viewer App

github.com/praxstack/markdown-viewer-app

Production markdown viewer — real-time preview, syntax highlighting, Mermaid diagrams, 10 themes, PDF/HTML export. Modular vanilla JS, sub-100 KB bundle.

Metrics 155+ Vitest tests · Mermaid · PDF/HTML export.

ORIGIN

EDUCATION

M.C.A., Computer Science · MNNIT Prayagraj GPA 9.25
2019 — 2022

B.Sc. (Hons), Computer Science · Bhaskaracharya College, University of Delhi
2015 — 2018

MILES

AWARDS

Amazon Pay Merchants Categories Champion (2023) — 100+ hotel trust-buster bugs; on-call 20+~5/shift.

THANK YOU FOR FLYING PRAXAIR · CLEARED FOR DEPARTURE · SR · SWE